Chapter 4: Network Access



Introduction to Networks

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Chapter 4

4.1 Physical Layer Protocols

4.2 Network Media

4.3 Data Link Layer Protocols

4.4 Media Access Control

4.5 Summary

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4.1 Physical Layer Protocols

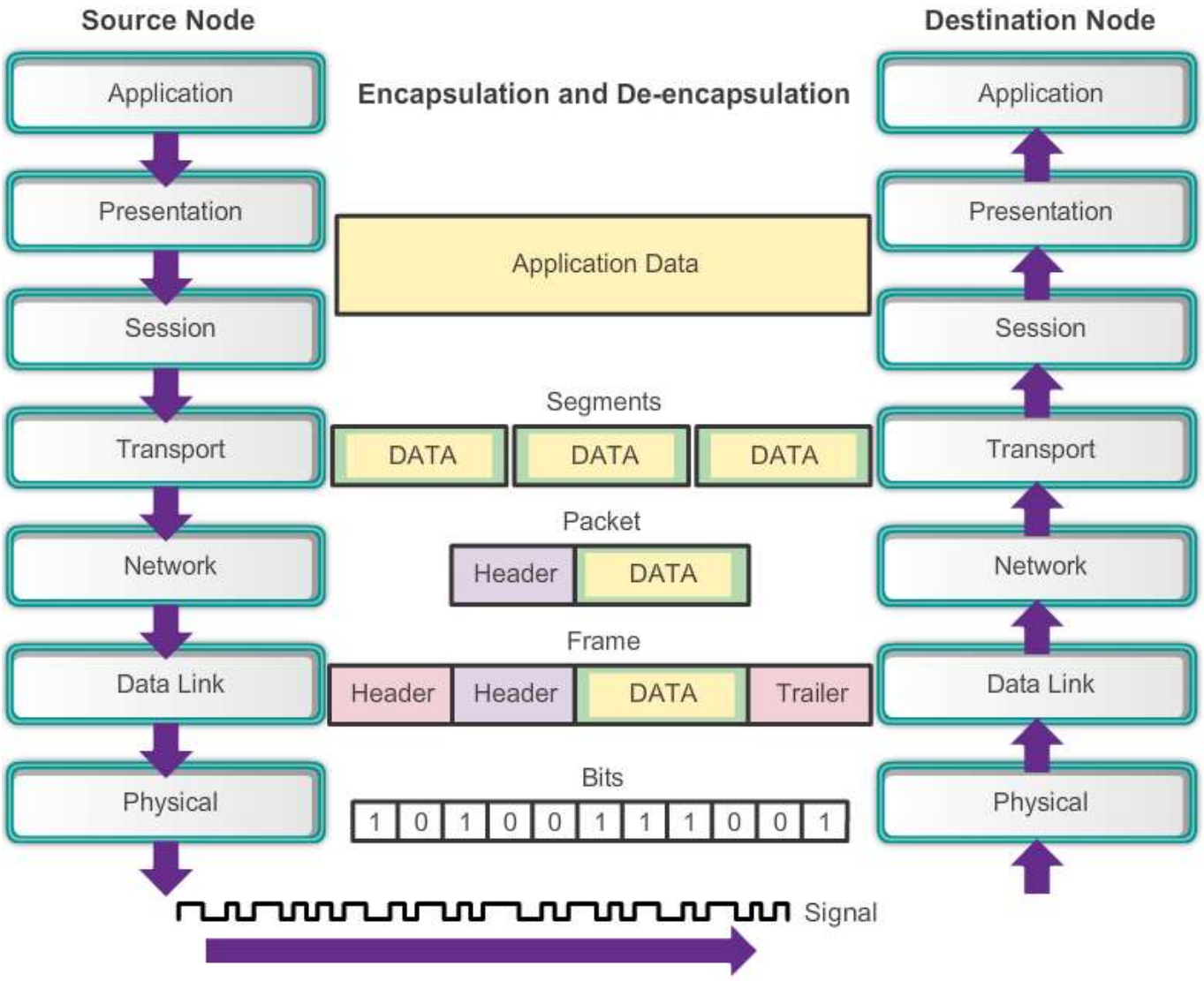


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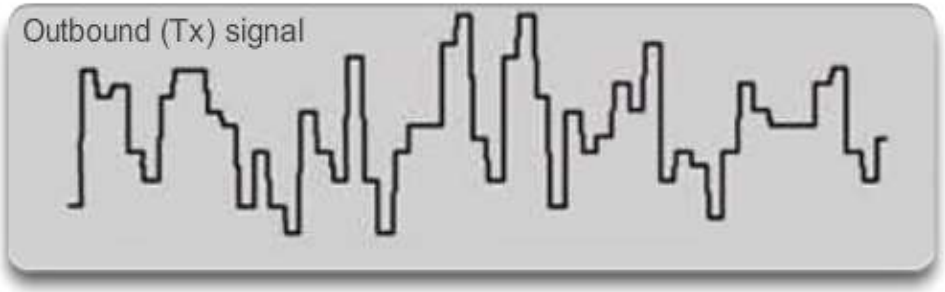
Purpose of the Physical Layer

The Physical Layer



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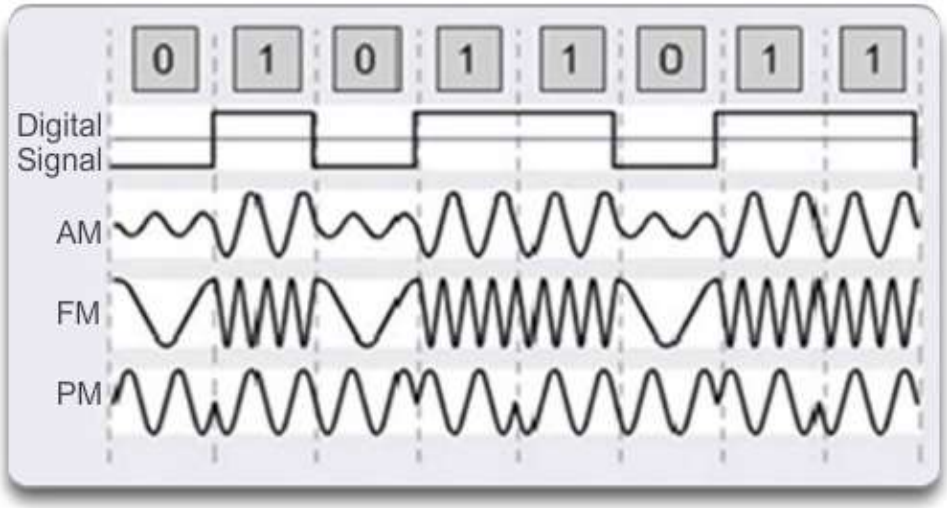
Physical Layer Media



Electrical Signals -
Copper cable



Light Pulse -
Fiber-optic cable



Microwave Signals -
Wireless

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Physical Layer Standards

Standard Organization	Networking Standards
ISO	• ISO 8877: Officially adopted the RJ connectors (e.g., RJ-11, RJ-45) • ISO 11801: Network cabling standard similar to EIA/TIA 568.
EIA/TIA	• TIA-568-C: Telecommunications cabling standards, used by nearly all voice, video and data networks. • TIA-569-B: Commercial Building Standards for Telecommunications Pathways and Spaces • TIA-598-C: Fiber optic color coding • TIA-942: Telecommunications Infrastructure Standard for Data Centers
ANSI	• 568-C: RJ-45 pinouts. Co-developed with EIA/TIA
ITU-T	• G.992: ADSL
IEEE	• 802.3: Ethernet • 802.11: Wireless LAN (WLAN) & Mesh (Wi-Fi certification) • 802.15: Bluetooth

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Bandwidth

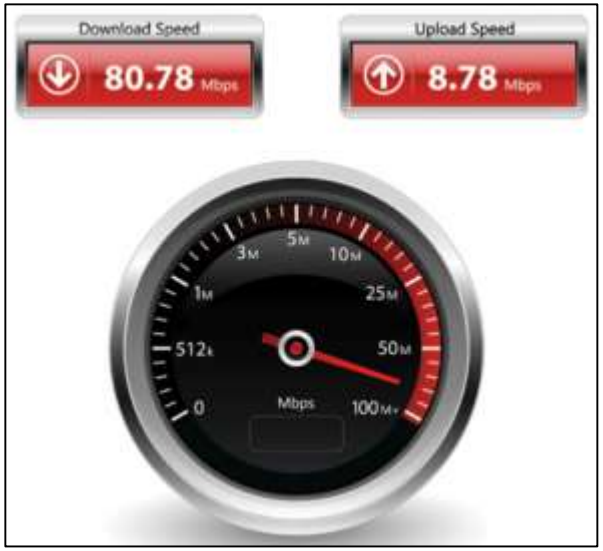
Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	kbps	1 kbps = 1,000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps = 1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^12 bps

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Throughput

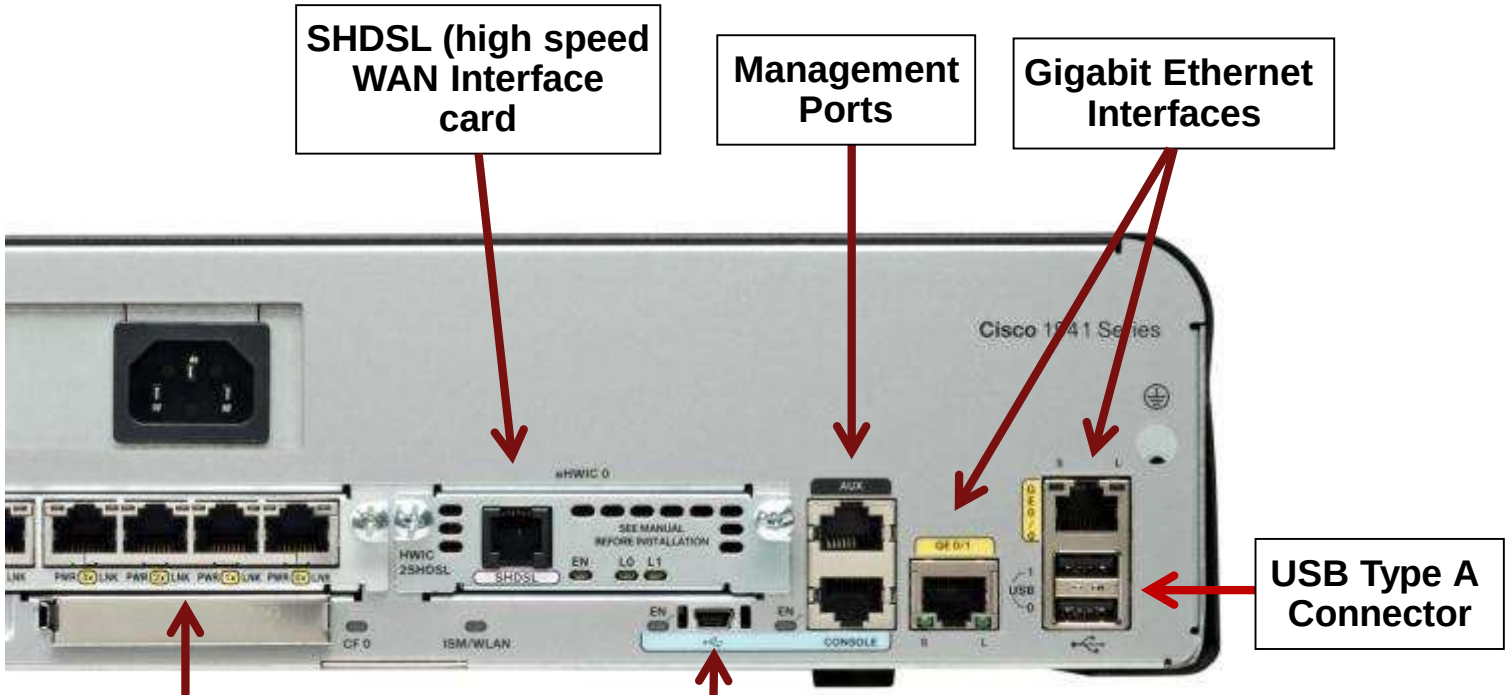
Throughput = the number of bits transferred across the media over a given period of time. Factors that can influence this transfer includes:

- The amount of traffic
- Type of traffic
- Latency/delays created by the number of network devices between source and destination



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Types of Physical Media



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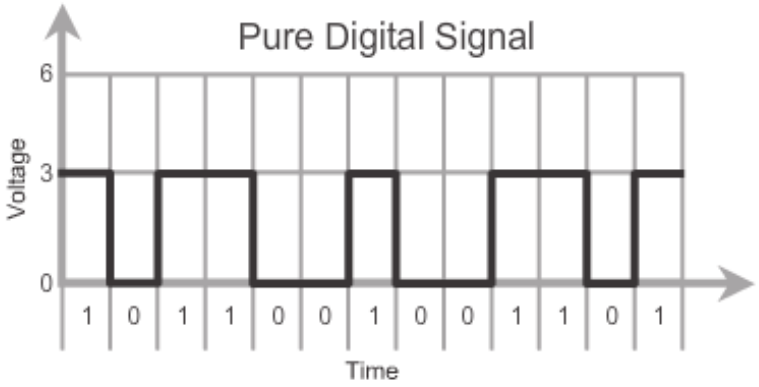
4.2 Network Media (cabling)

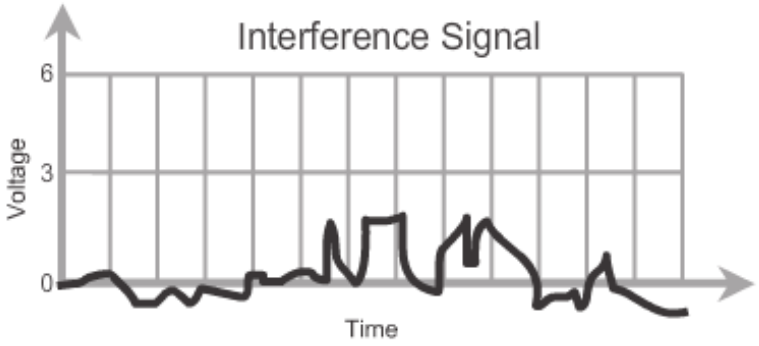


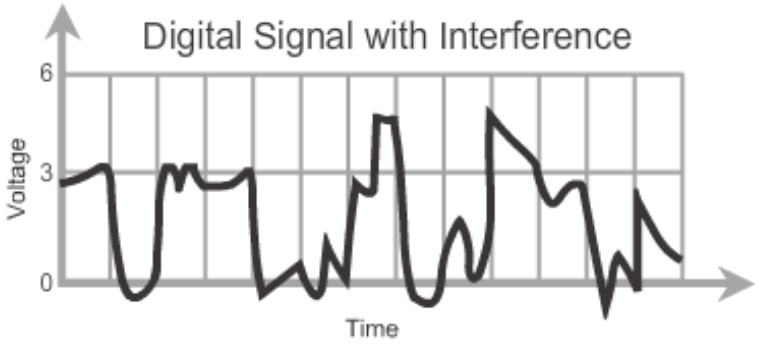
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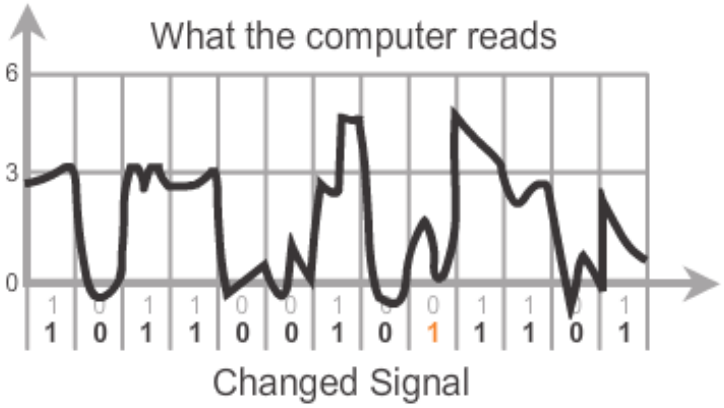
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Characteristics of Copper Media



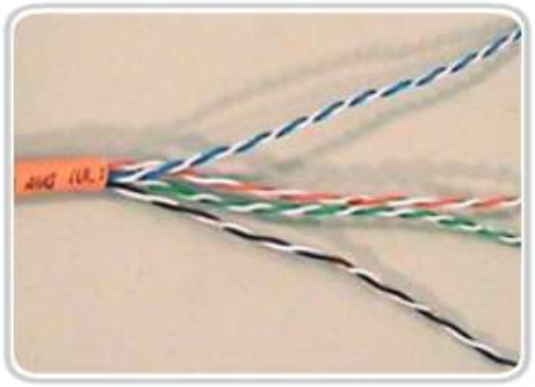






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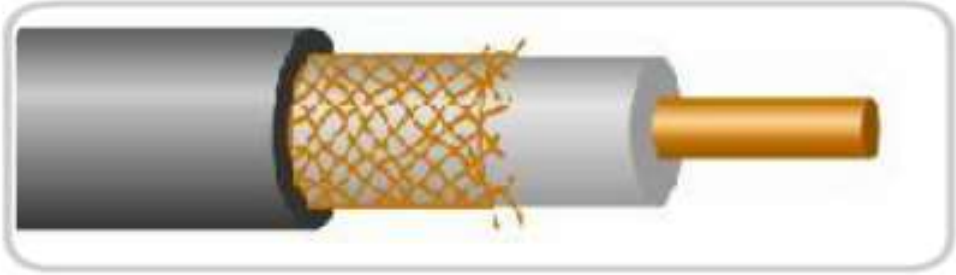
Copper Media



Unshielded Twisted Pair (UTP) Cable



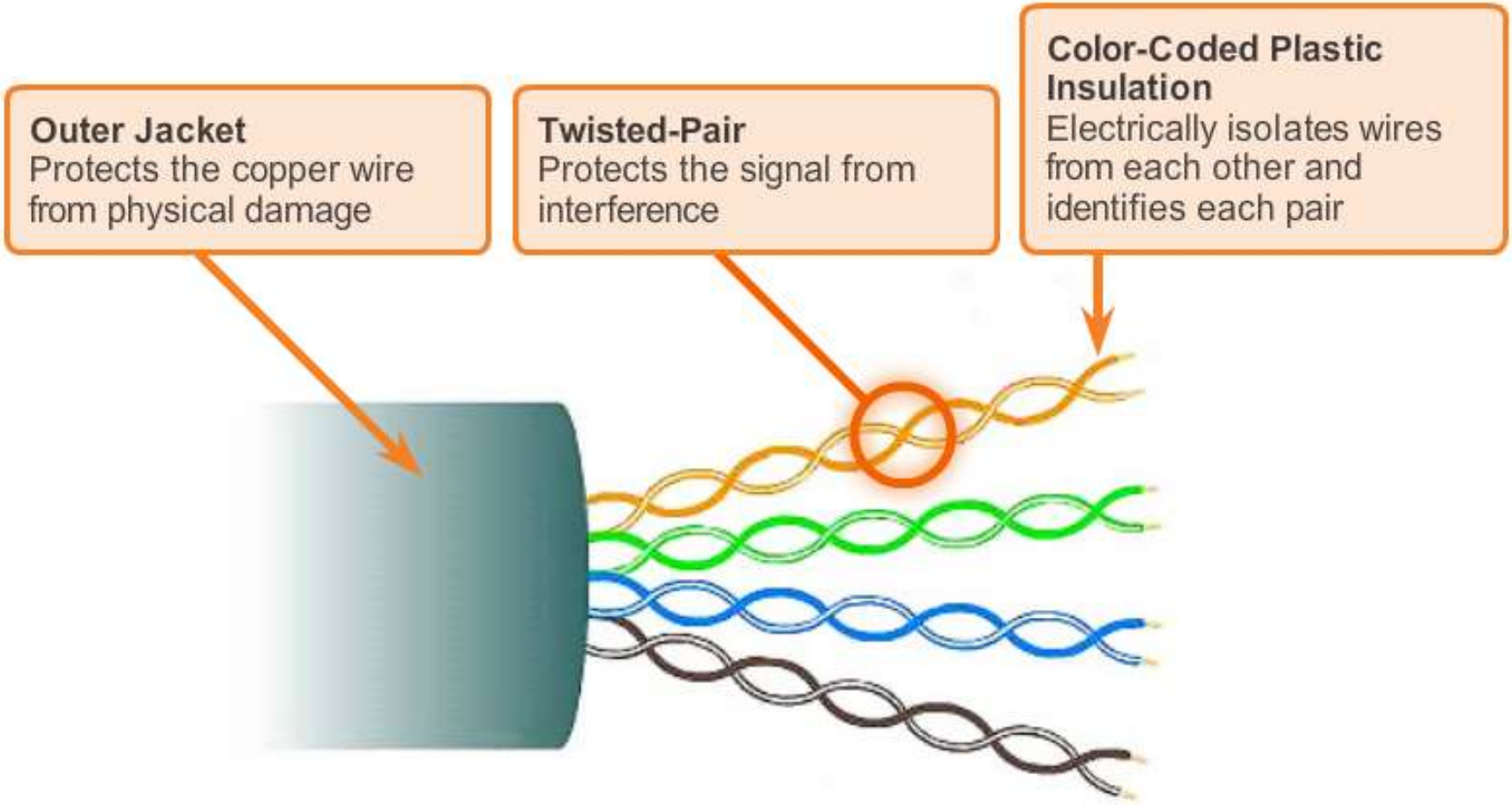
Shielded Twisted Pair (STP) Cable



Coaxial Cable

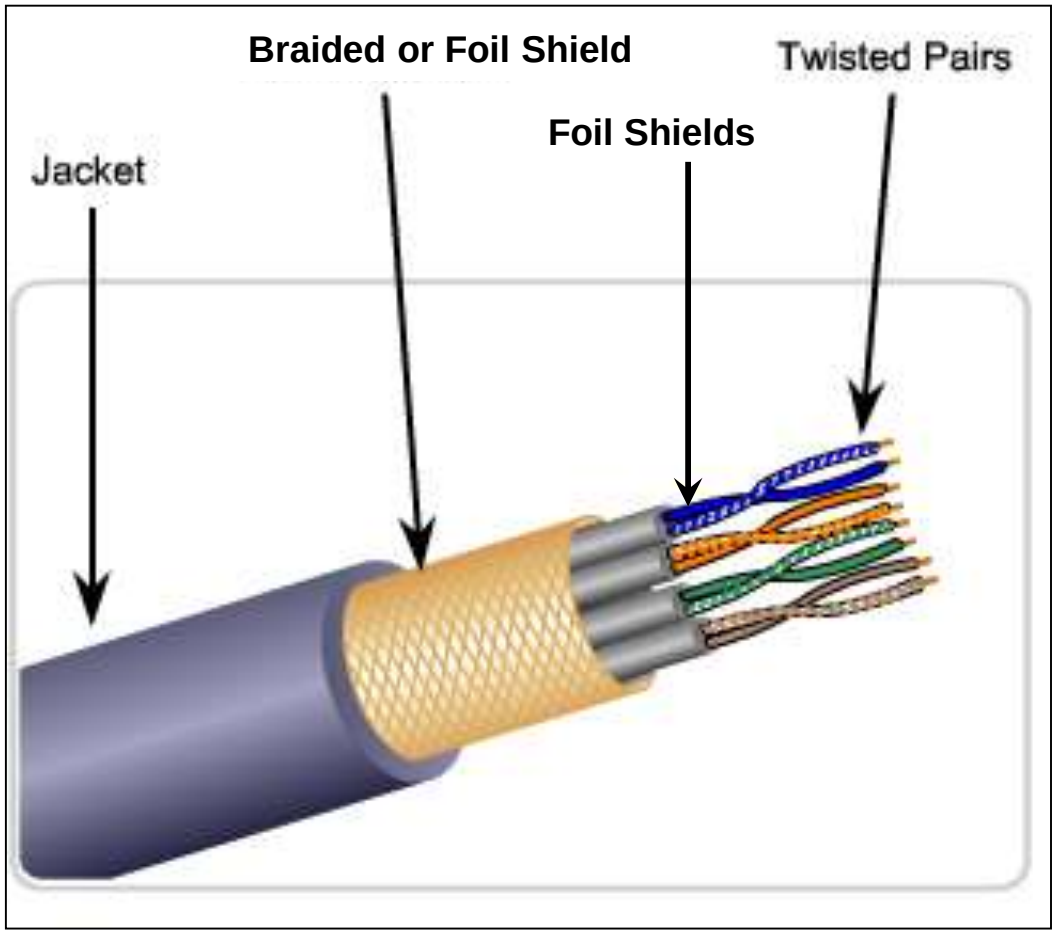
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UTP Cable



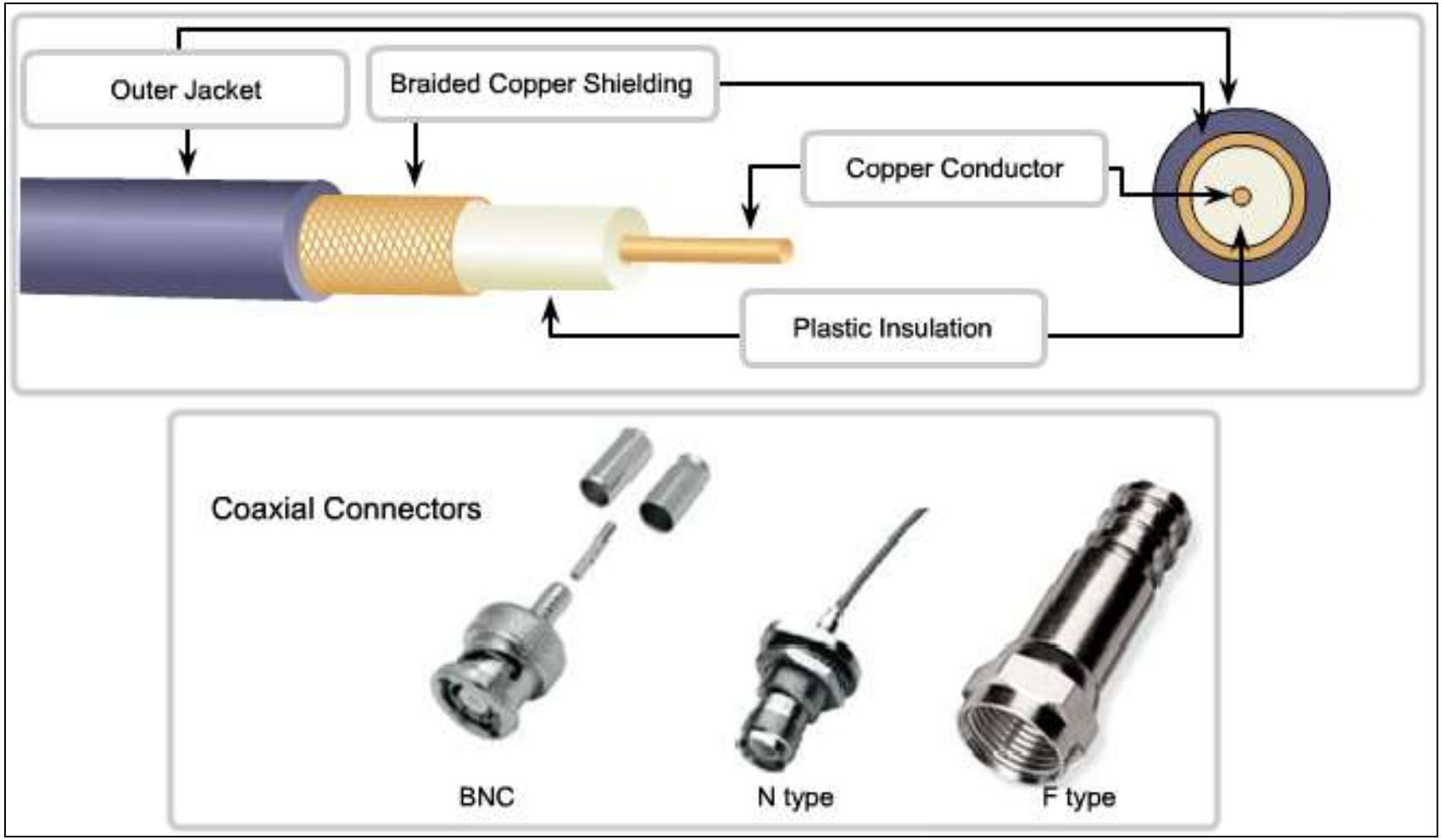
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STP Cable



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Coaxial Cable

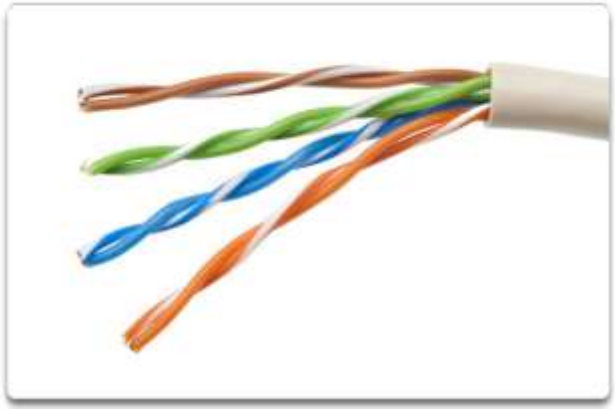


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Properties of UTP Cabling

UTP cable does not use shielding to counter the effects of EMI (electric magnetic interference) and RFI (radio frequency interference). Instead, cable designers have discovered that they can limit the negative effect of crosstalk by:

- Cancellation: when two wires in an electrical circuit are **placed close together**, their magnetic fields are the exact opposite of each other. Therefore, the two magnetic fields cancel each other out and also cancel out any outside EMI and RFI signals.
- Varying the number of **twists** per wire pair: UTP cable must follow precise specifications governing how many twists or braids are permitted per meter (3.28 feet) of cable.
- The orange/orange white pairs are twisted less than the blue/white blue pair.

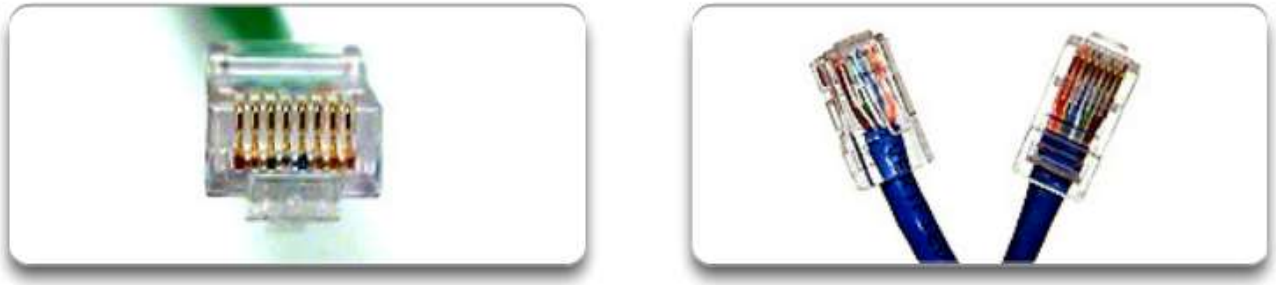


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UTP Connectors

Each time copper cabling is terminated there is the possibility of signal loss and the introduction of noise to the communication circuit.

RJ-45 UTP Plugs



RJ-45 UTP Socket



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Types of UTP Cable

Cable Type	Standard	Application
Ethernet Straight-through	Both ends T568A or both ends T568B	Connects a network host to a network device such as a switch or hub.
Ethernet Crossover	One end T568A, other end T568B	• Connects two network hosts • Connects two network intermediary devices (switch to switch, or router to router)
Rollover	Cisco proprietary	Connects a workstation serial port to a router console port, using an adapter.

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Testing UTP Cables

After installation, a UTP cable tester should be used to test for the following parameters:

- Wire map (proper pin termination at each end, shorts etc)
- Cable length
- Signal loss due to attenuation (loss of signal strength)
- Crosstalk

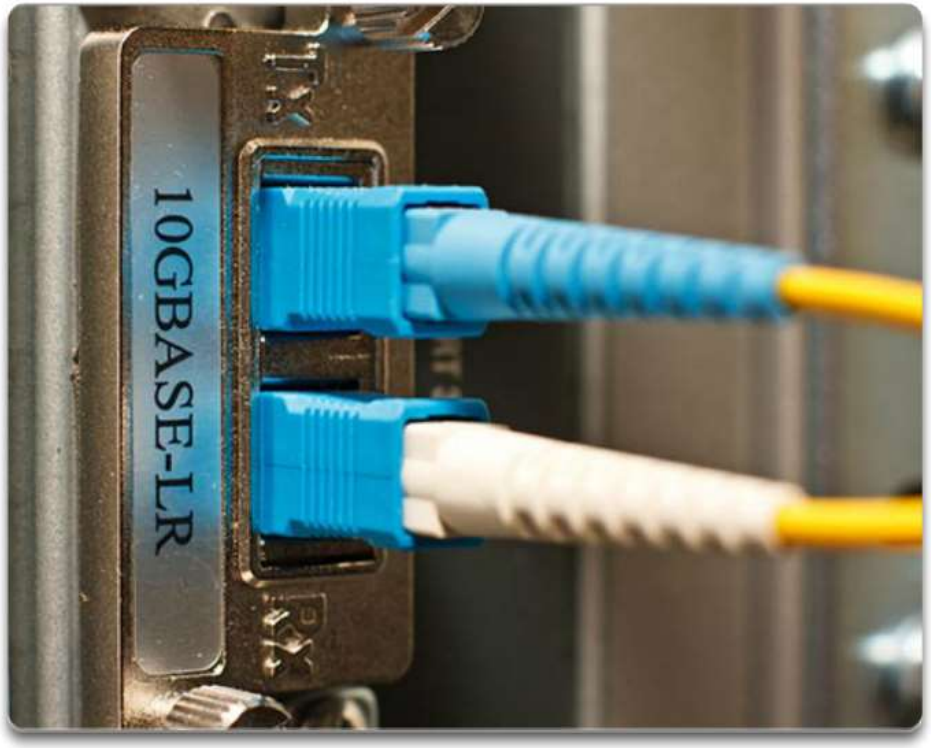


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Properties of Fiber Optic Cabling

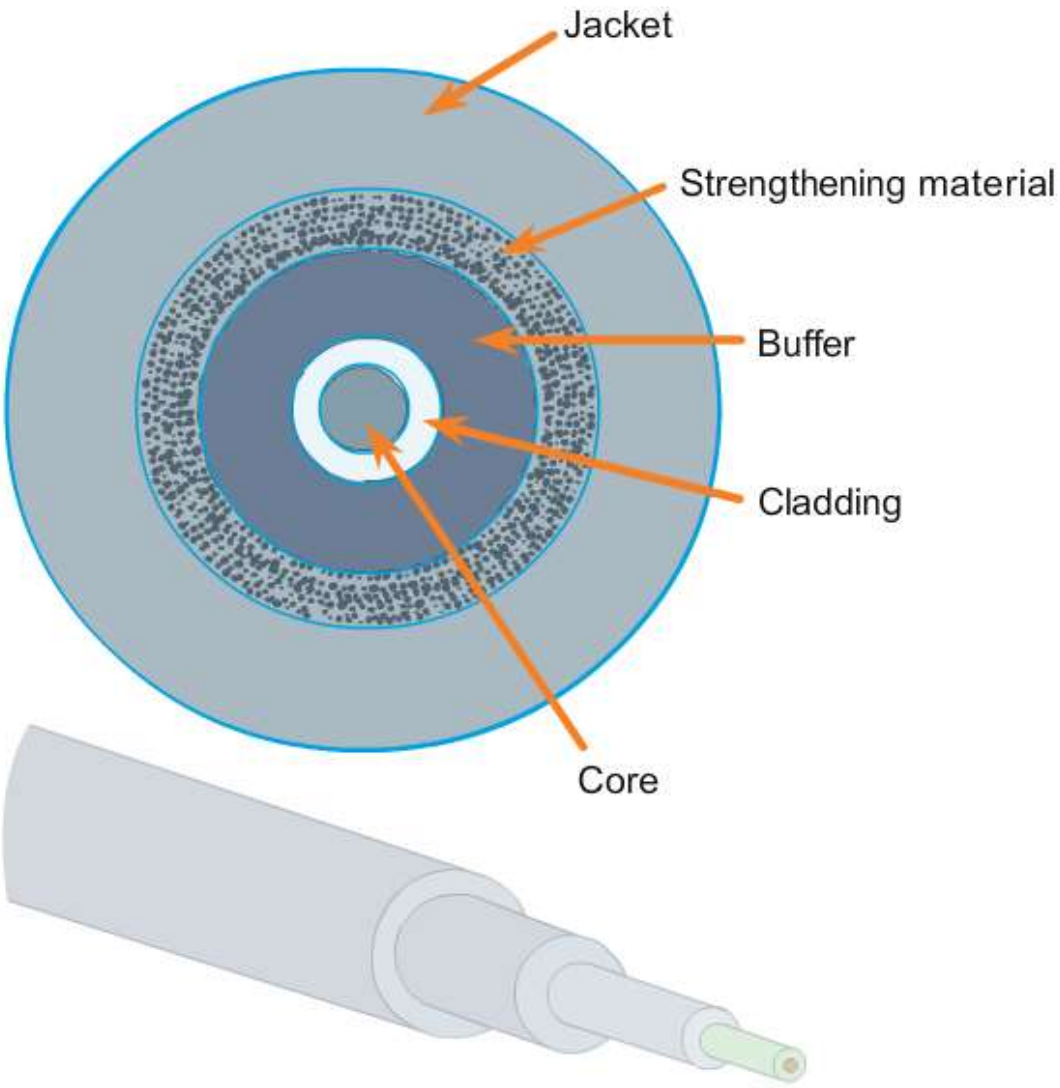
Fiber-optic cabling is now being used in four types of industry:

- Enterprise Networks (backbone)
- Fiber-to-the-home (FTTH) and Access Networks (always-on broadband services)
- Long-Haul Networks (ISP use it to connect to countries)
- Submarine Networks (harsh under sea environments)



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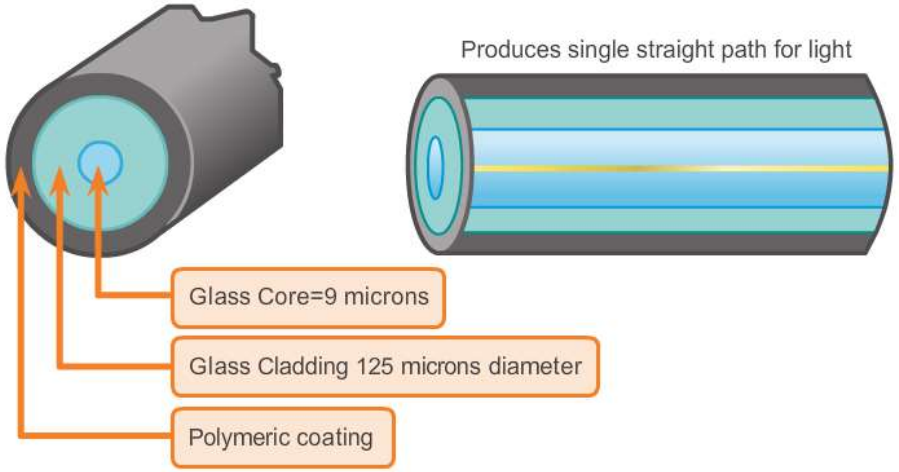
Fiber Media Cable Design



Fiber Optic Cabling

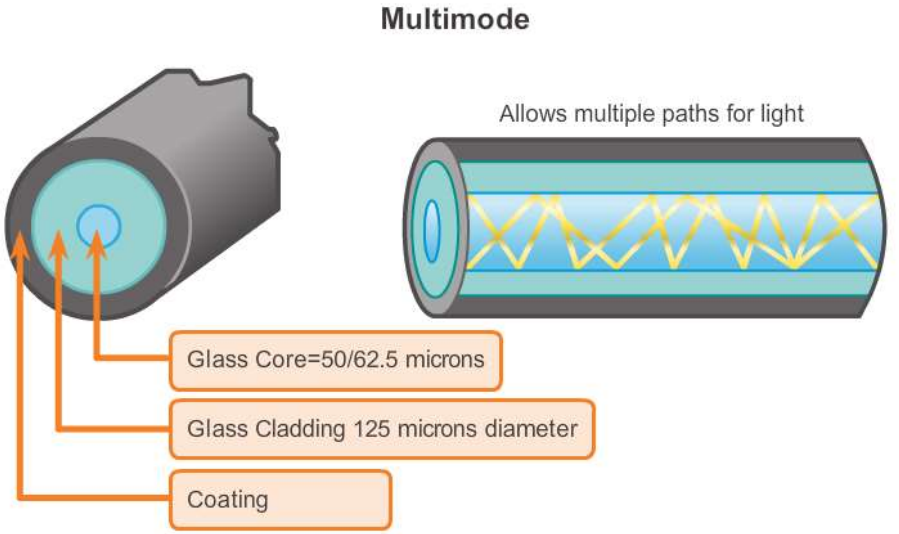
Types of Fiber Media

Single Mode



- Small core
- Less dispersion
- Suited for long distance applications
- Uses lasers as the light source
- Commonly used with campus backbones for distances of several thousand meters

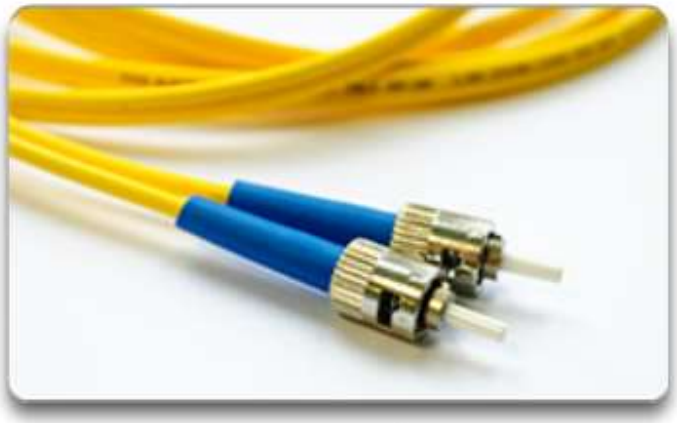
Multimode



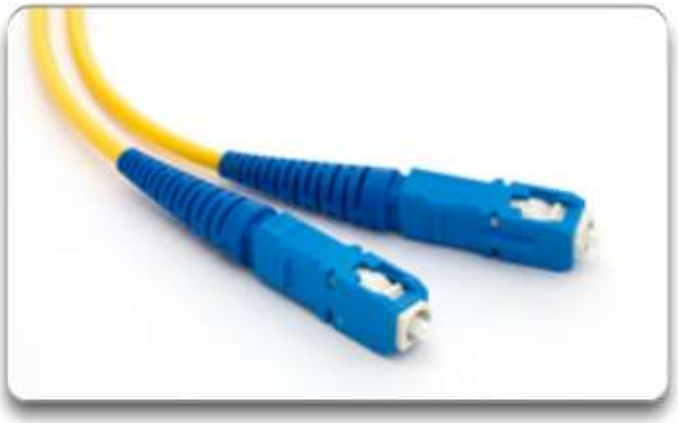
- Larger core than single mode cable
- Allows greater dispersion and therefore, loss of signal
- Suited for long distance applications, but shorter than single mode
- Uses LEDs as the light source
- Commonly used with LANs or distances of a couple hundred meters within a campus network

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Network Fiber Connectors



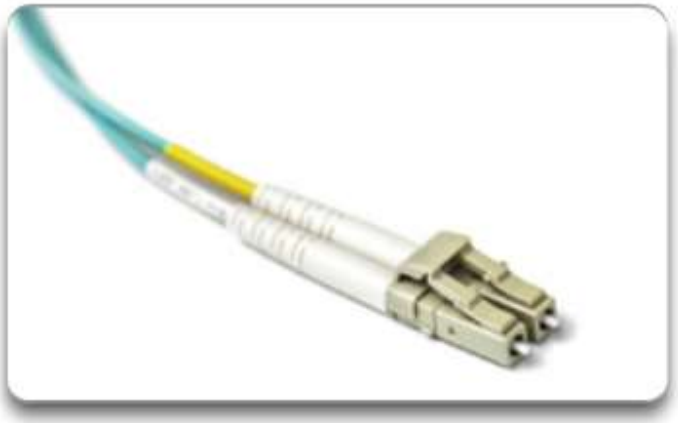
ST Connectors



SC Connectors



LC Connector



Duplex Multimode LC Connectors

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Fiber versus Copper

Implementation Issues	Copper Media	Fibre Optic
Bandwidth Supported	10 Mbps – 10 Gbps	10 Mbps – 100 Gbps
Distance	Relatively short (1 – 100 meters)	Relatively High (1 – 100,000 meters)
Immunity To EMI And RFI	Low	High (Completely immune)
Immunity To Electrical Hazards	Low	High (Completely immune)
Media And Connector Costs	Lowest	Highest
Installation Skills Required	Lowest	Highest
Safety Precautions	Lowest	Highest

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Properties of Wireless Media

Wireless does have some areas of concern including:

- Coverage area
- Interference
- Security



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Types of Wireless Media

	• IEEE 802.11 standards • Commonly referred to as Wi-Fi. • Uses CSMA/CA • Variations include: • 802.11a: 54 Mbps, 5 GHz - business • 802.11b: 11 Mbps, 2.4 GHz - home • 802.11g: 54 Mbps, 2.4 GHz – compatible with b • 802.11n: 600 Mbps, 2.4 and 5 GHz • 802.11ac: 1 Gbps, 5 GHz • 802.11ad: 7 Gbps, 2.4 GHz, 5 GHz, and 60 GHz
	• IEEE 802.15 standard • Supports speeds up to 3 Mb/s • Provides device pairing over distances from 1 to 100 meters.
	• IEEE 802.16 standard • Provides speeds up to 1 Gbps • Uses a point-to-multipoint topology to provide wireless broadband access.

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4.3 Data Link Layer Protocols

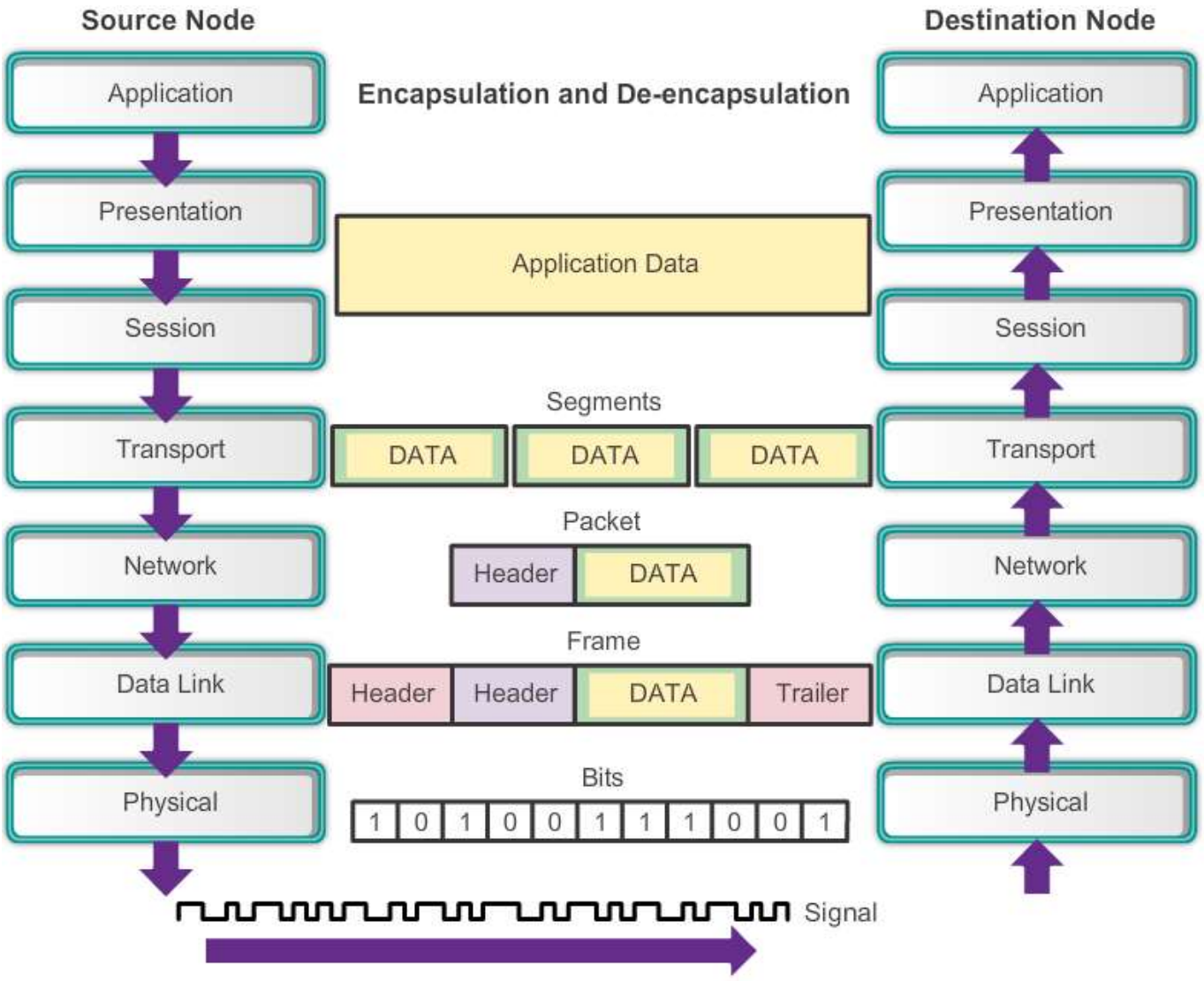


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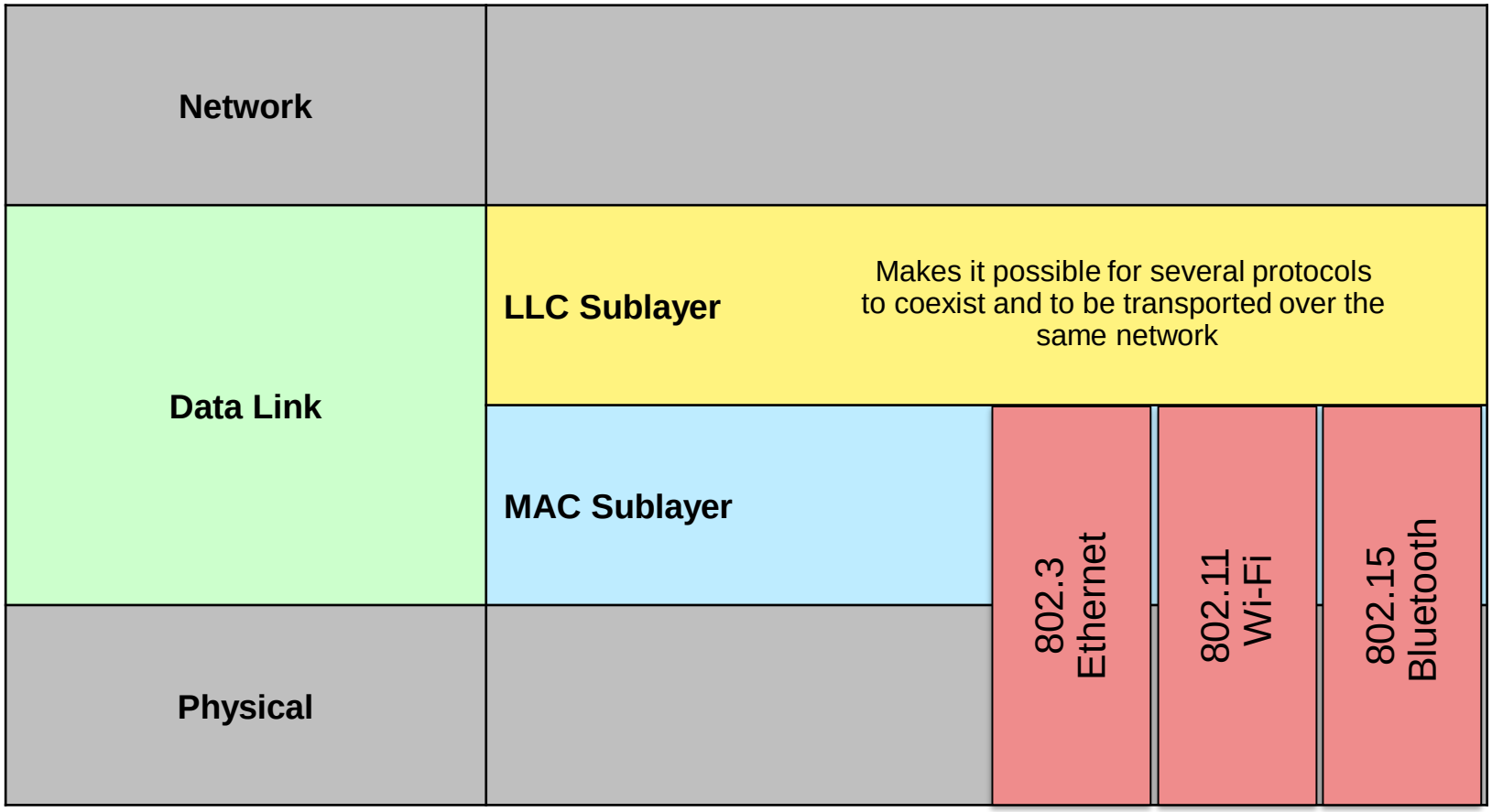
Purpose of the Data Link Layer

The Data Link Layer



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Data Link Sublayers



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Media Access Control

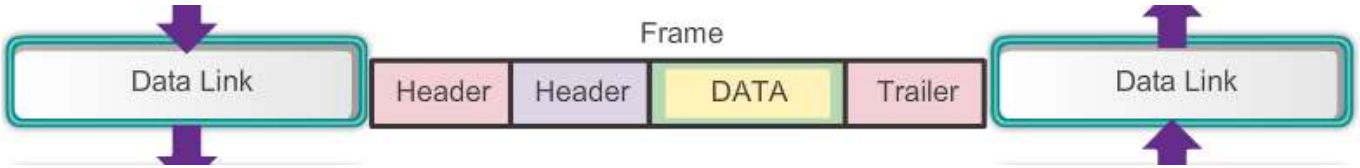
The Data Link Layer

Data link layer protocols govern how to format a frame for use on different media.

Different protocols may be in use for different media.

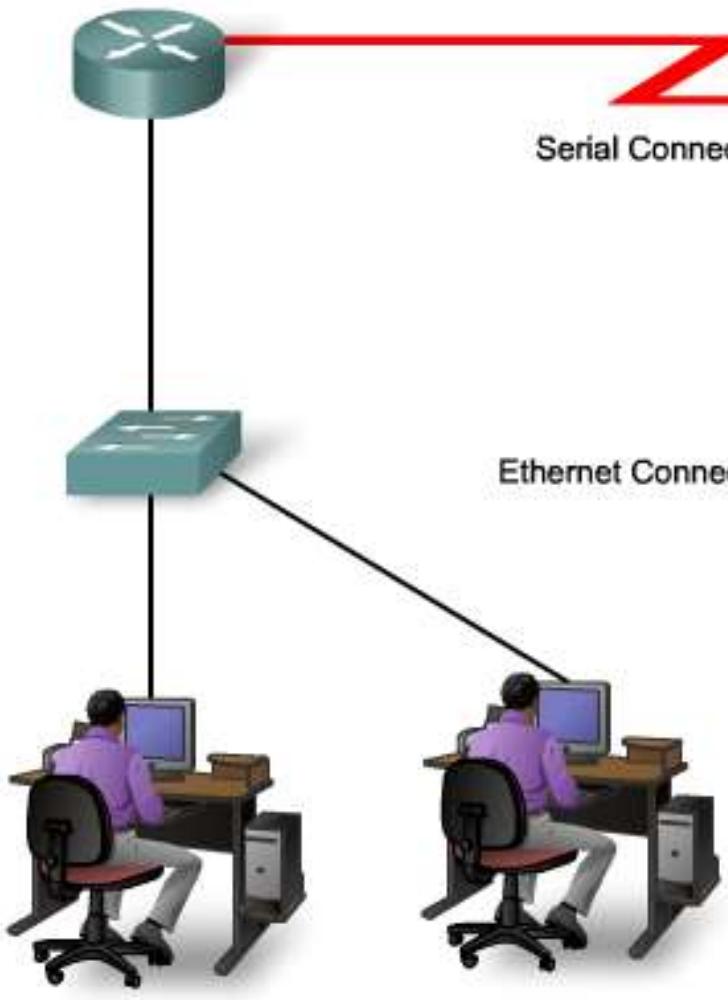


At each hop along the path, an intermediary device accepts frames from one medium, decapsulates the frame and then forwards the packets in a new frame. The headers of each frame are formatted for the specific medium that it will cross.



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Providing Access to Media



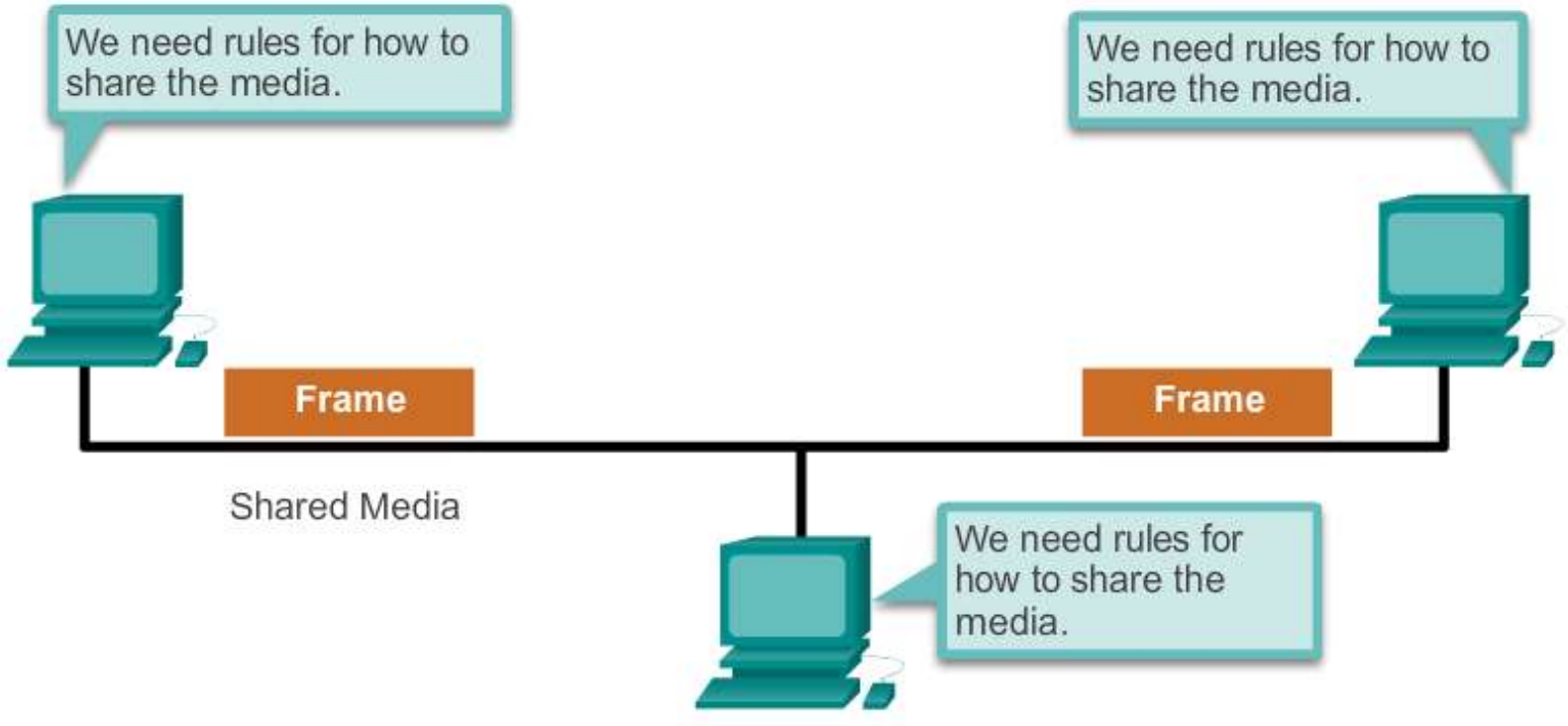
The Data Link layer is responsible for controlling the transfer of frames across the media.

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Media Access Control

Media Access Control method depends on:

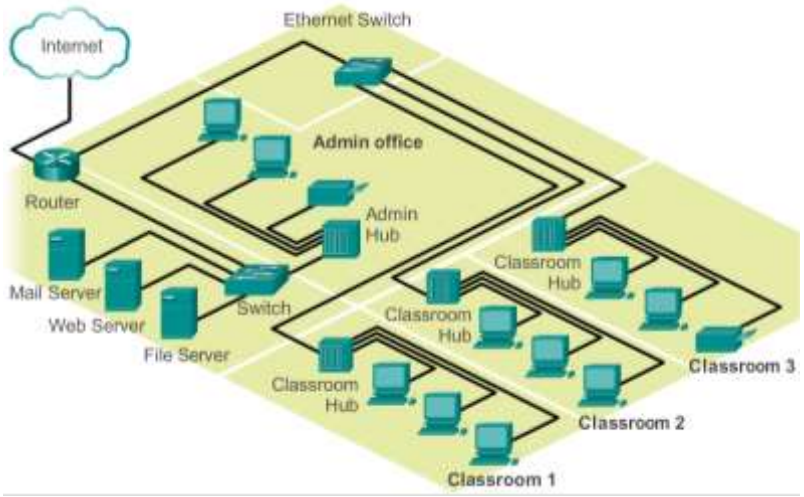
- **Topology**
- Media sharing



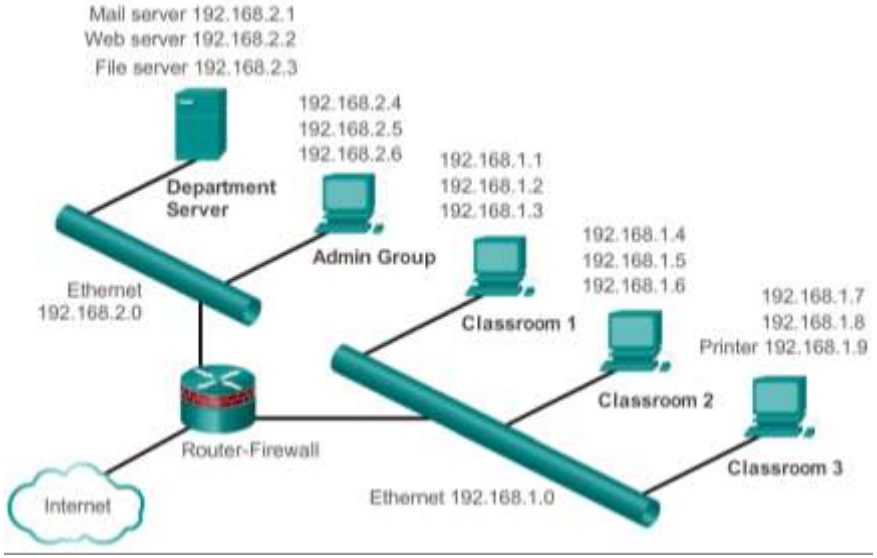
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Physical and Logical Topologies

Physical **Topology** : star or point-to-point

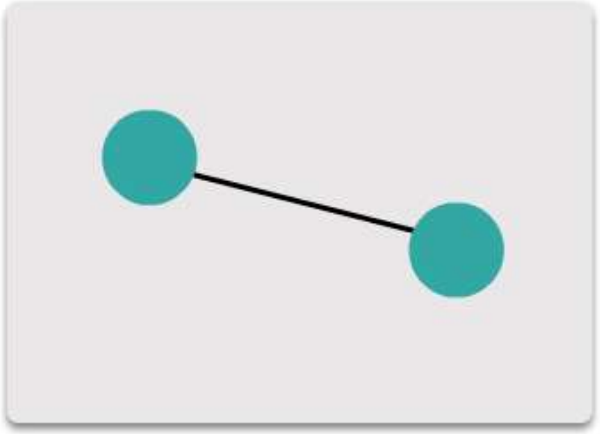


Logical **Topology**

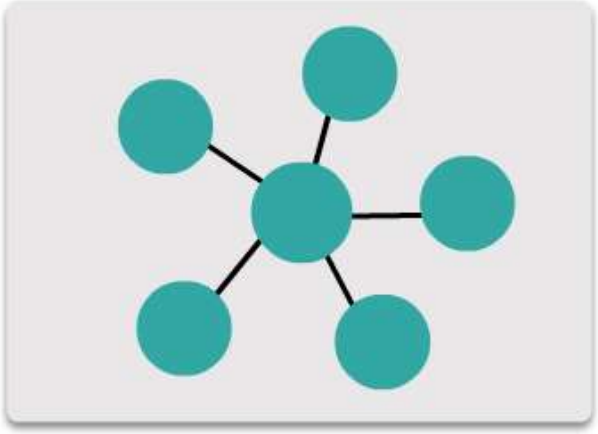


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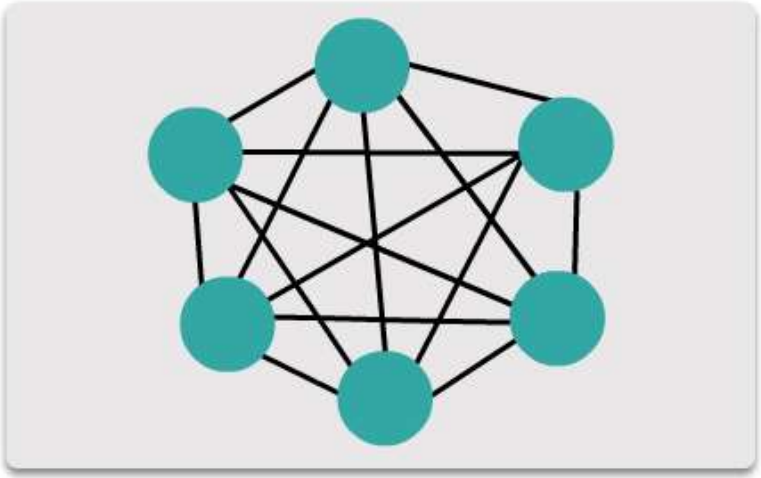
Common Physical WAN Topologies



Point-to-point topology



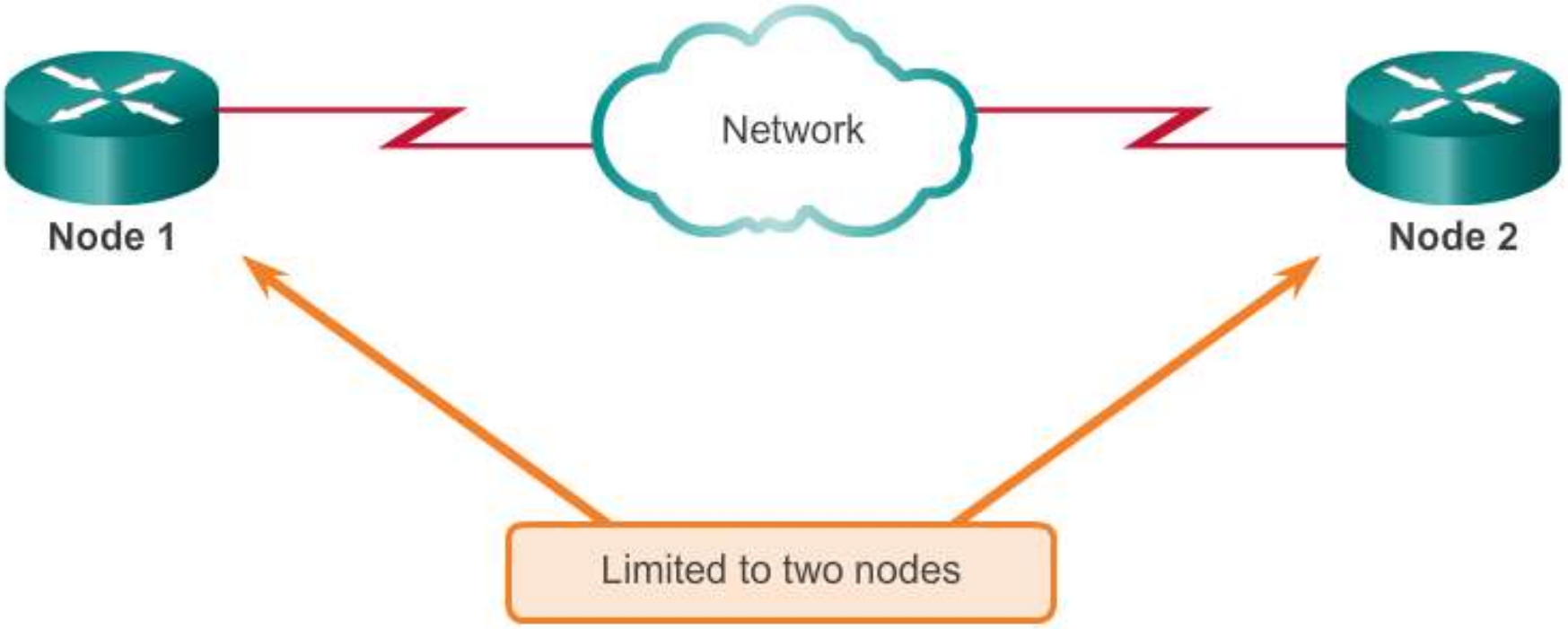
Hub and spoke topology



Full mesh topology

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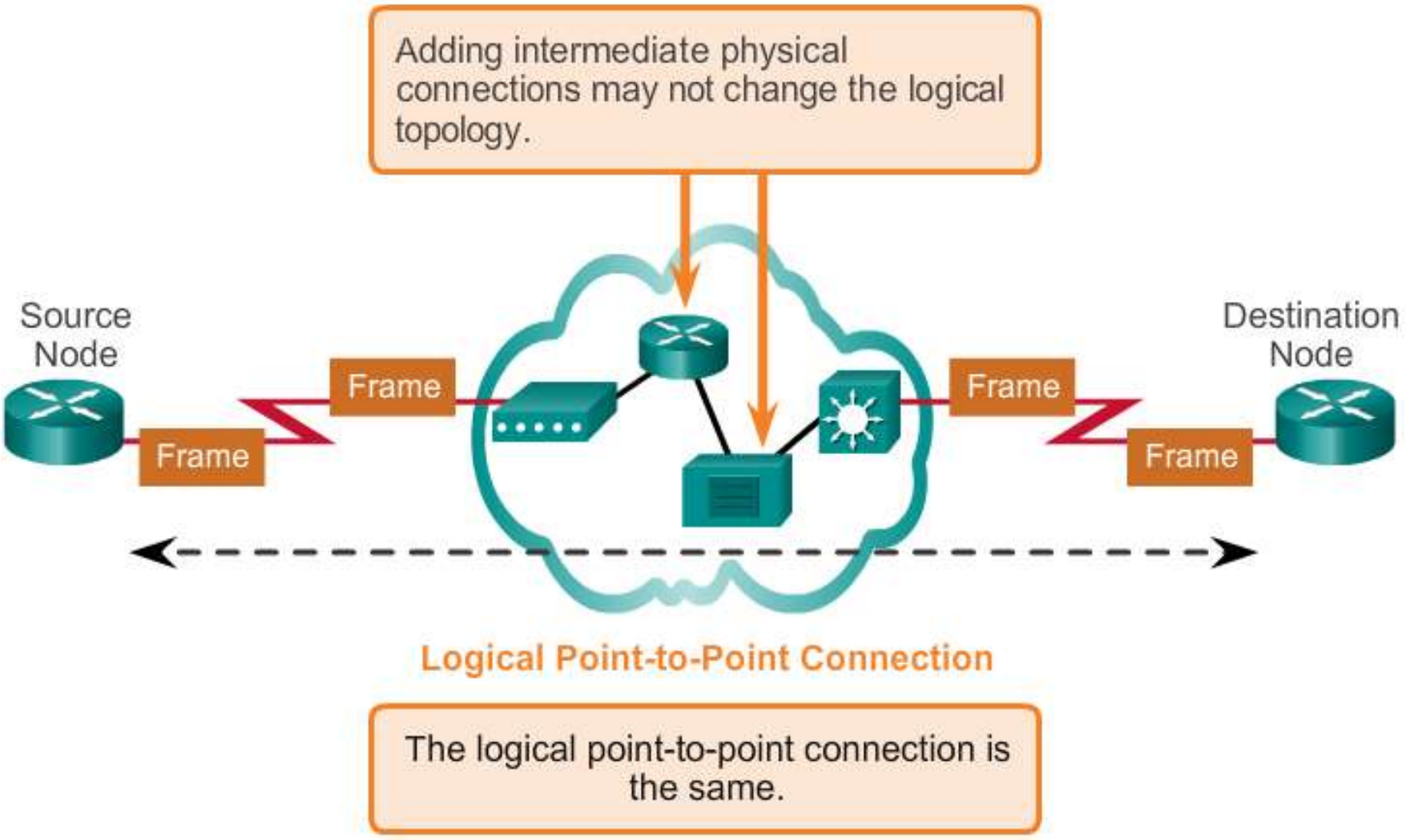
Physical Point-to-Point Topology



The router is the only one to use this media

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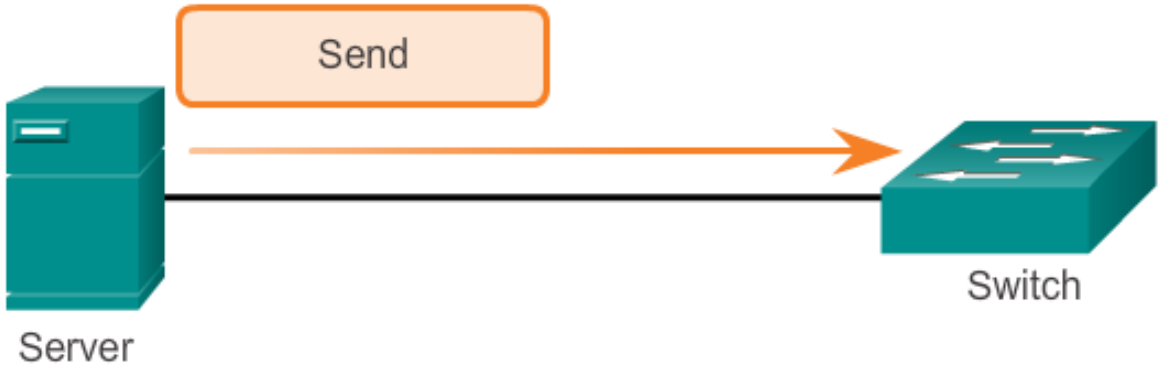
Logical Point-to-Point Topology



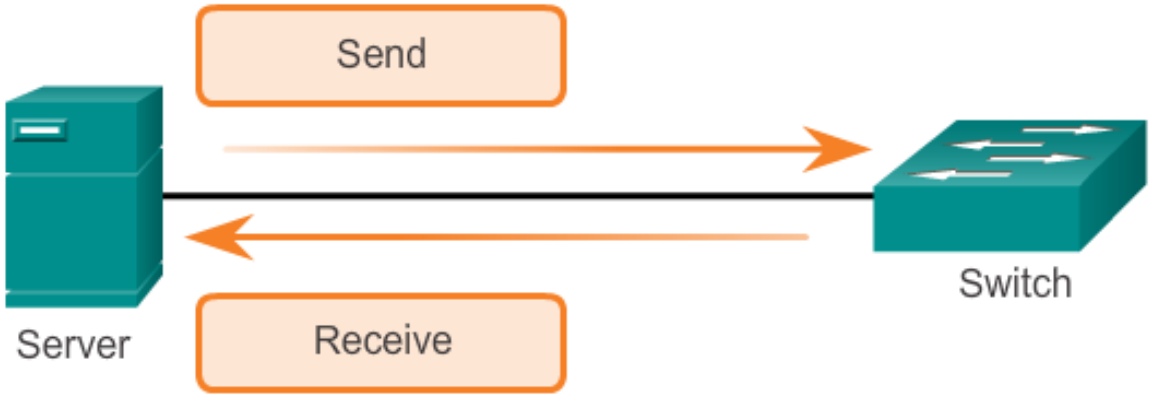
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Half- and Full-Duplex

Half-Duplex

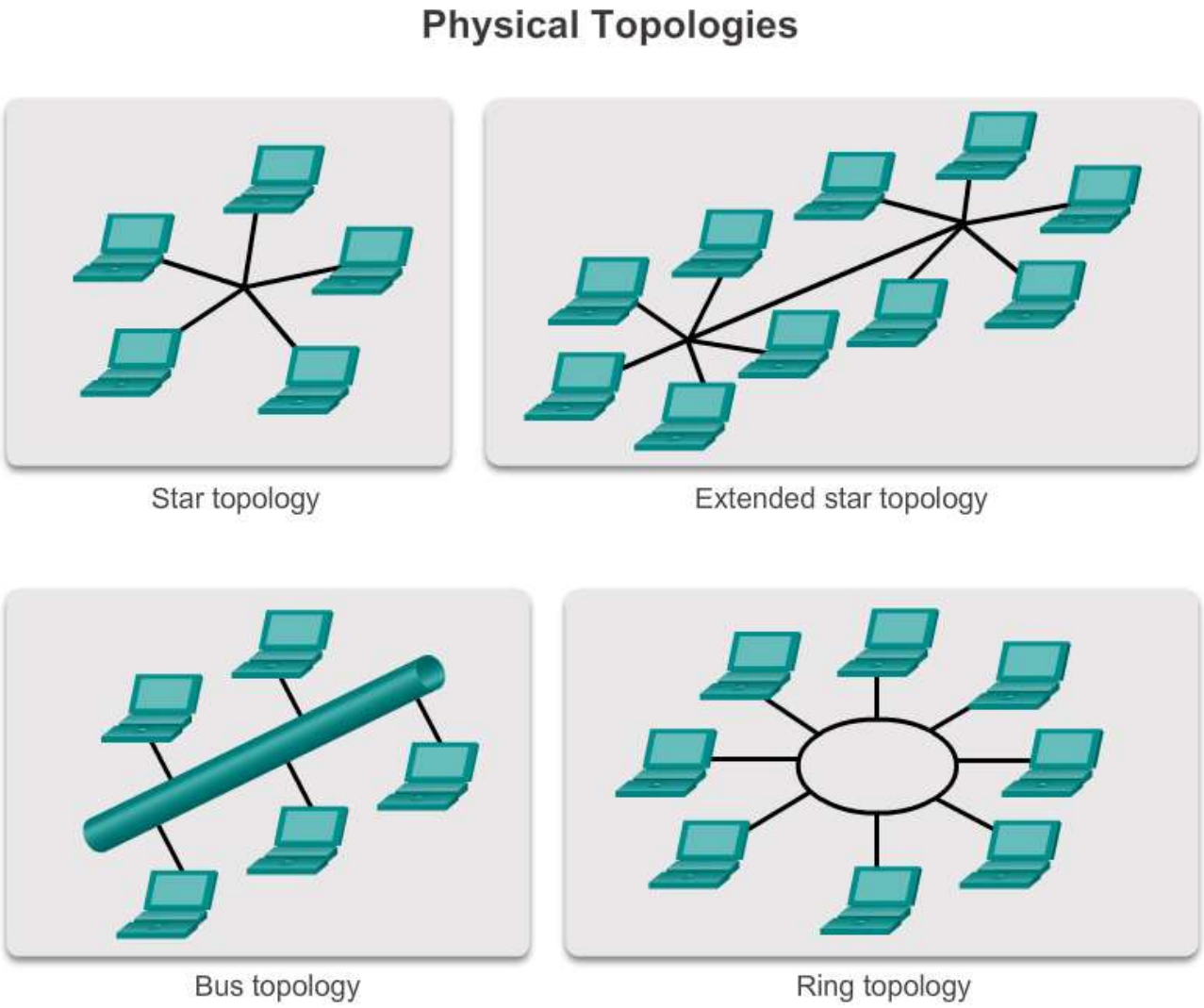


Full-Duplex



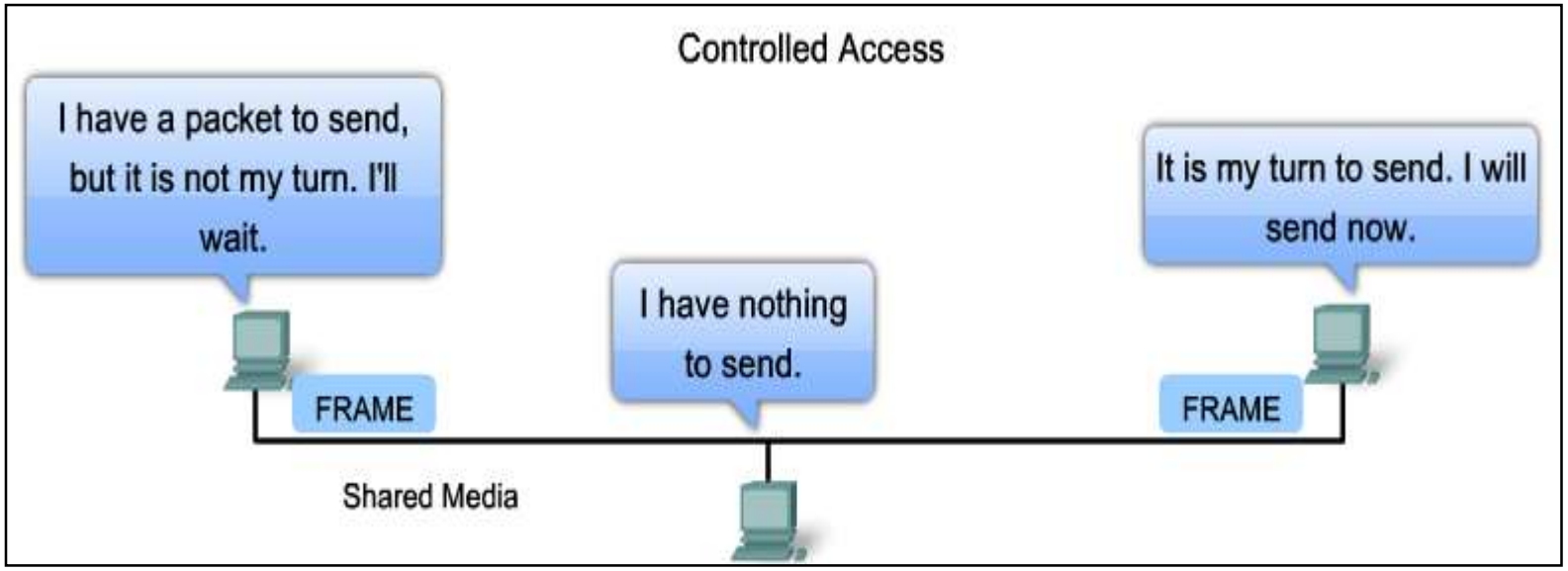
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Physical LAN Topologies



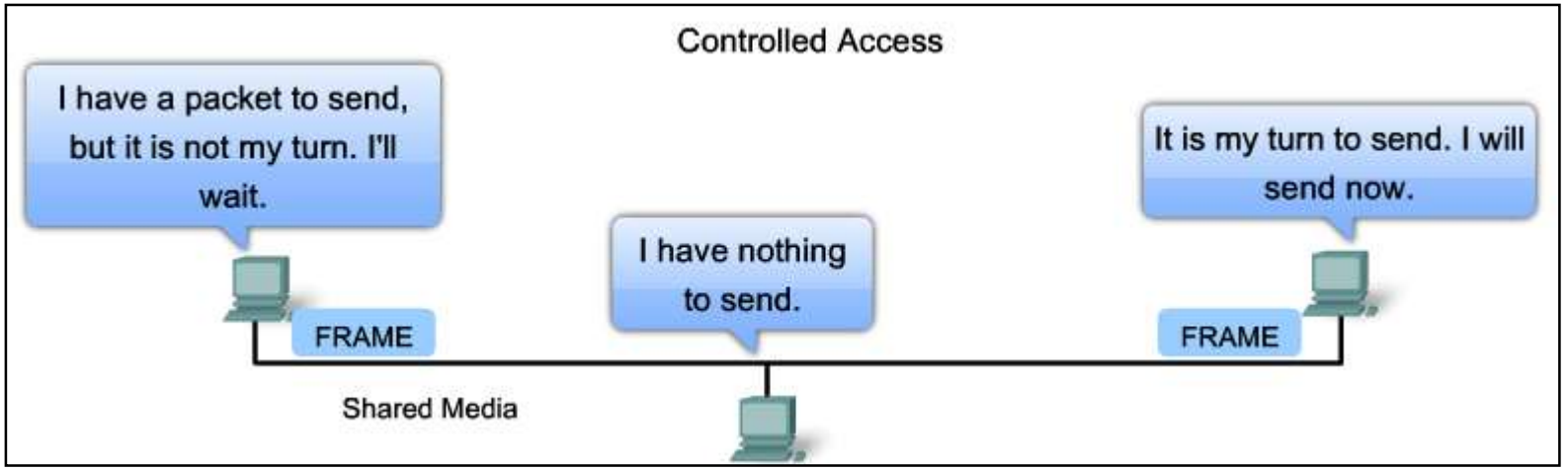
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Logical Topology for Shared Media



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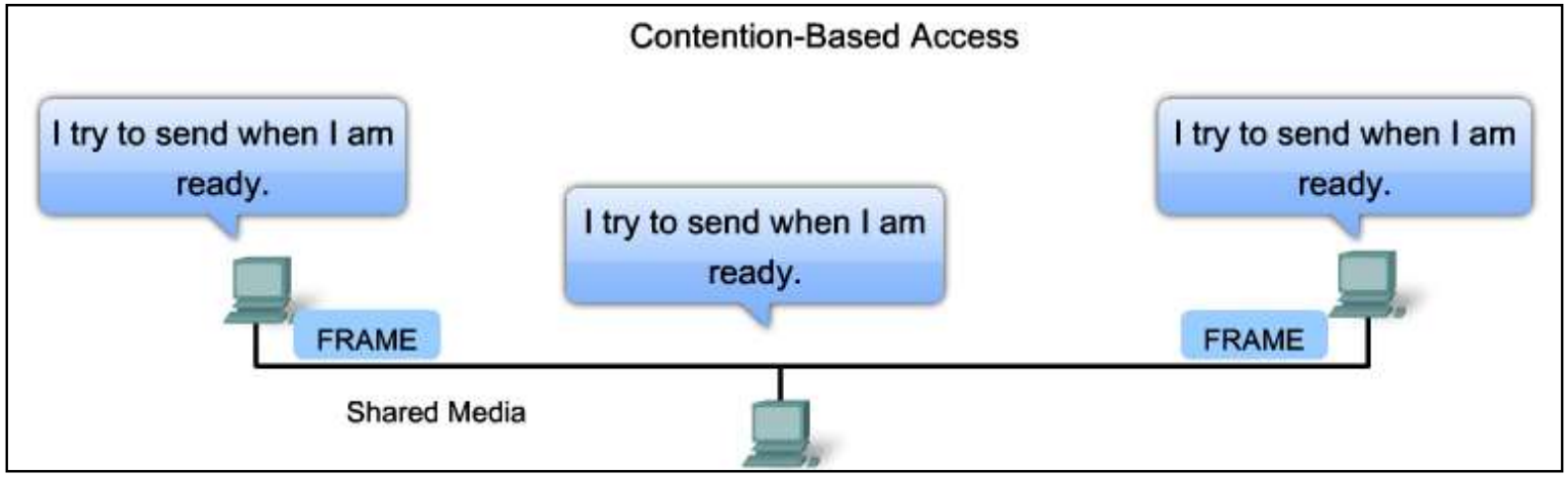
Controlled Access



Characteristics	Controlled Access Technologies
• Only one station can transmit at a time • Devices wanting to transmit must wait their turn • No collisions • May use a token passing method	• Token Ring (IEEE 802.5) • FDDI

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Contention-Based Access

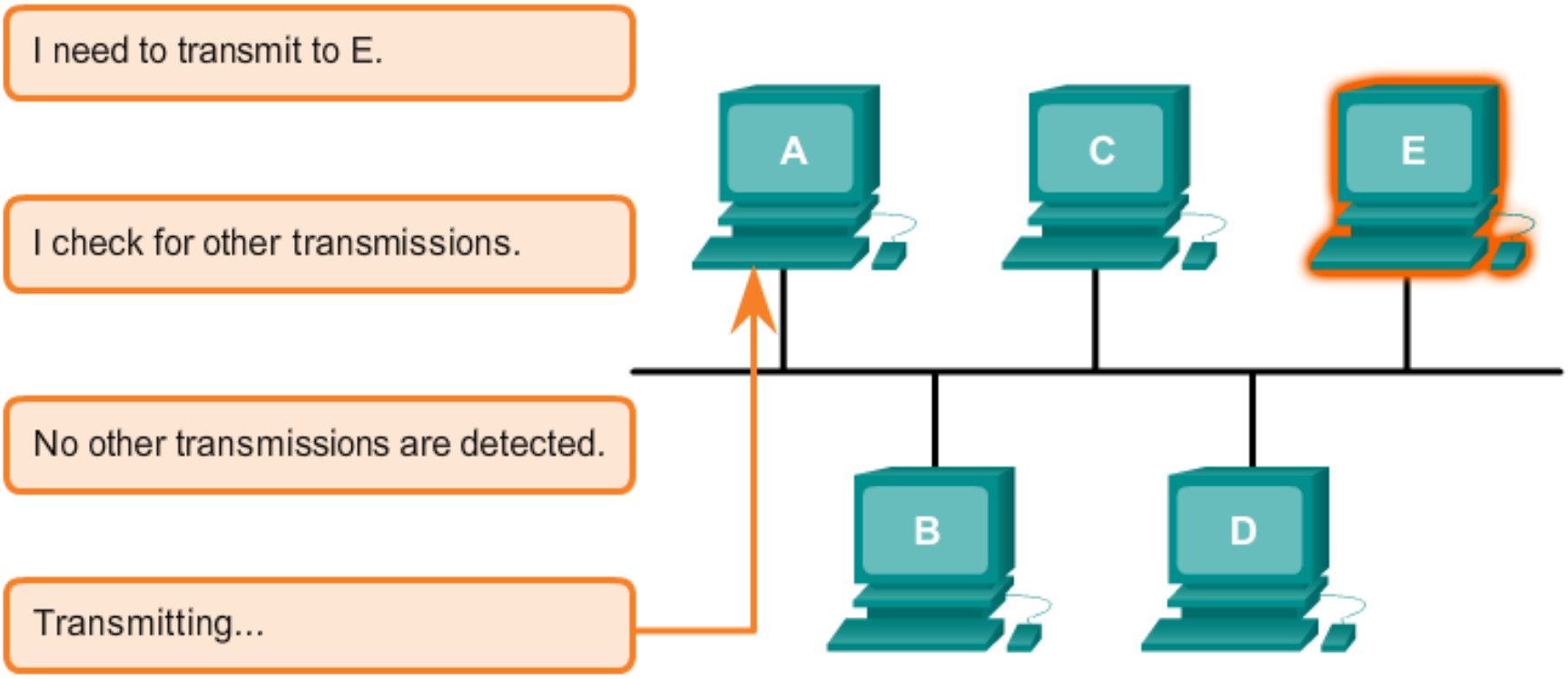


Characteristics	Contention-Based Technologies
• Stations can transmit at any time • Collision exist • There are mechanisms to resolve contention for the media	• CSMA/CD for 802.3 Ethernet networks • CSMA/CA for 802.11 wireless networks

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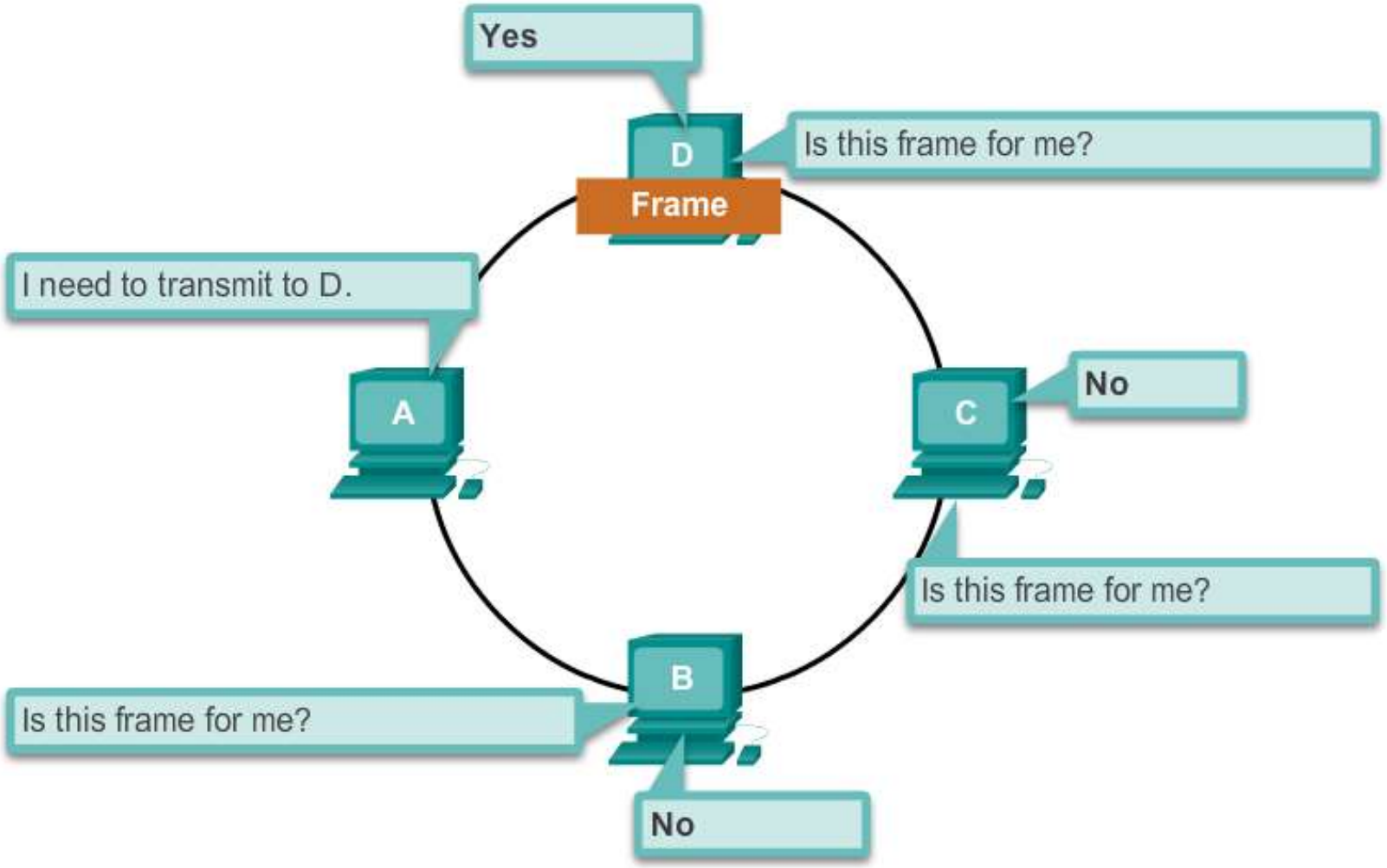
Multi-Access Topology

Logical Multi-Access Topology



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Ring Topology



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The Frame

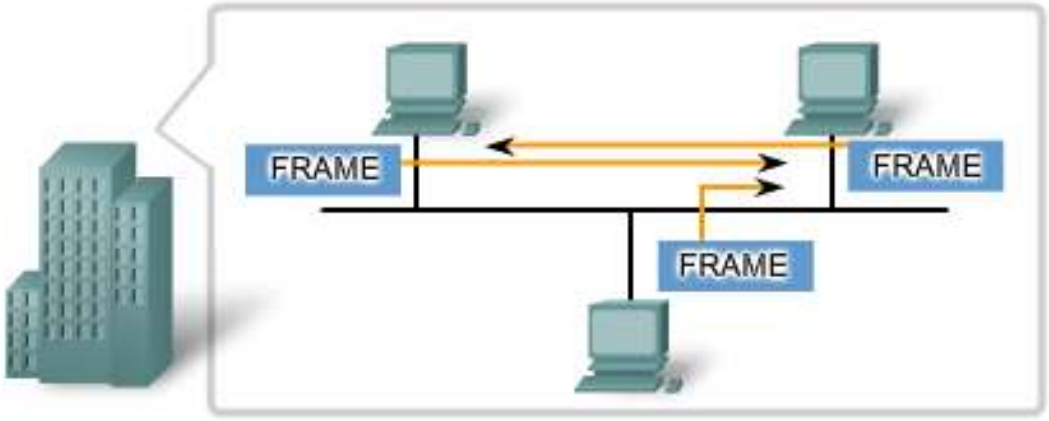
In a fragile environment, more controls are needed to ensure delivery. The header and trailer fields are larger as more control information is needed.

Greater effort needed to ensure delivery = higher overhead = slower transmission rates



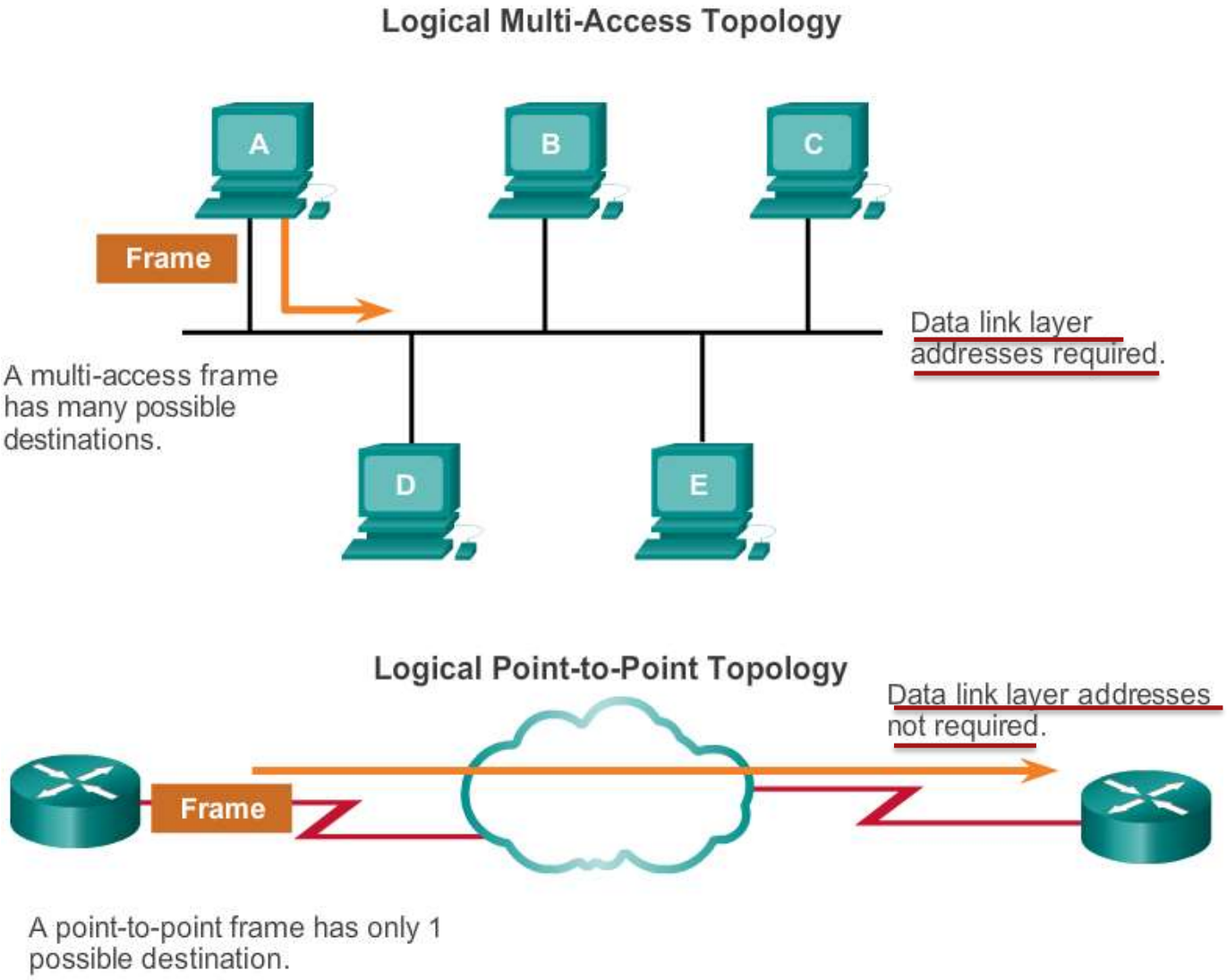
In a protected environment, we can count on the frame arriving at its destination. Fewer controls are needed, resulting in smaller fields and smaller frames.

Less effort needed to ensure delivery = lower overhead = faster transmission rates



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Layer 2 Address



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Summary

- The TCP/IP network access layer is the equivalent of the OSI data link layer (Layer 2) and the physical layer (Layer 1).
- The OSI physical layer provides the means to transport the bits that make up a data link layer frame across the network media.
- The physical layer standards address three functional areas: physical components, frame encoding technique, and signaling method.
- Using the proper media is an important part of network communications. Without the proper physical connection, either wired or wireless, communications between any two devices will not occur.
- Wired communication consists of copper media and fiber cable.
- There are three main types of copper media used in networking: unshielded-twisted pair (UTP), shielded-twisted pair (STP), and coaxial cable. UTP cabling is the most common copper networking media.

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